Zev-7 & Metronome

Nixie & Numitron Clocks

Operator's Guide

September 2026

This manual describes the installation, configuration, and day-to-day operation of the Zev-7 and Metronome clocks. It covers the Nixie-tube models (Zev-7, Zev-7 Mini, Zev-7-18, Zev-7RZ, Metronome) and the numitron-tube models (Zev-7NS, Zev-7ND). Throughout this manual all models are referred to as Zev-7, and "Nixie" refers to the time display in general; see Section 2 for how the numitron models differ.

Designed by Mitchell Feig

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Quick Start

Follow these steps to set up the clock for the first time. A full explanation of every option is in the numbered sections that follow.

1. Power on

Connect the clock to its 12V DC power adapter. On a brand-new clock, the display shows a single lit tube bouncing left and right across NX1 through NX7 (Zev-7) or NX1 through NX15 (Metronome), with the digit counting 0–9 as it bounces — this is AP configuration mode. On a previously-used clock that already has WiFi credentials, skip to step 5 — to reconfigure WiFi, hold SW2 for 5 seconds to enter AP mode. To factory reset all settings, hold SW1 and SW2 and apply power. Hold buttons until the display starts.

2. Display configuration

> **NOTICE:** If the display shows 00, the processor must be returned to the manufacturer for reprogramming.

Several clock variants are available. The processor is shipped programmed for each specific configuration.

3. Connect to the clock's WiFi

With the AP-mode bouncing tube visible, from a phone or computer, open the WiFi settings and connect to the network named for your model — Zev-7S, Zev-7M (Mini), Zev-7-18, Zev-7RZ, Zev-7NS or Zev-7ND (numitron), or Metronome. No password is required.

4. Configure in a browser

Open a web browser and go to 192.168.4.1. The setup page loads. At minimum:

- On the Network page, select the home WiFi network from the SSID list and enter the password.
- On the Time Zone page, confirm the UTC offset for the local region (US Eastern is 5 hours Minus by default).

Tap Save, then tap Back, then tap Start. The clock saves the settings and reboots.

5. Wait for time to appear

During boot, NX1 briefly shows a scrolling digit "1" while WiFi is connecting, then a scrolling digit "2" while the clock fetches time from the internet. The time appears within a few seconds of connecting to WiFi.

6. Daily use

To do this	Do this
Change brightness temporarily	Click SW2 to cycle through brightness levels 1–9
Wake the clock when a rule has turned the display off	Click SW1
See the clock's IP address	Double-click SW1 — the address scrolls across the tubes
Adjust any setting live	Browse to http://<ip-address>/ from any device on the same WiFi
Install new firmware over WiFi	Hold SW1 for 10 seconds

7. Manual cathode conditioning (Nixie models only)

To recondition the Nixie tubes after shipping or storage, run a one-time 5-minute conditioning cycle: remove power, hold SW2, then reapply power while keeping SW2 held. Release once the digits start cycling. The clock will rapidly cycle every digit on every tube at full brightness for 5 minutes, then continue with normal startup. See Section 14 for full details.

8. Recommended next steps

- Visit the live web interface to explore LED effects, neon patterns, and digit change animations.
- Configure Rules on the live page to dim or turn off the display during sleeping hours, schedule LED lighting changes throughout the day, and disable the motion sensor at chosen times.
- If the optional weather sensor is installed, enable Temperature, Pressure, and Humidity on the Sensors page.

1. Overview

The Zev-7NS and Zev-7ND are numitron-tube variants of the Zev-7. They run their own firmware — functionally similar to the Zev-7's, differing where the display hardware does — and their seven displays are numitron seven-segment filament tubes driven by TLC59116 constant-current drivers rather than high-voltage Nixie tubes. Because numitron filaments do not suffer cathode poisoning, they have no anti-poison feature and no high-voltage circuitry. See Section 2 for the full list of differences.

Zev-7 is a seven-tube Nixie clock and the Metronome is its fifteen-tube companion. Both are driven by an ESP32 microcontroller and present the same web interface and rule-based scheduling, each running its own firmware. They display hours, minutes, seconds, and tenths of a second using Nixie tubes with HV5530 shift-register drivers. The Zev-7S can be built with any of several tube types; the other models each use one. The clocks feature RGB backlighting behind the tubes, neon colon indicators, WiFi connectivity with NTP time sync, an optional motion sensor, an optional light sensor, an optional weather sensor, and an under-light strip.

The Metronome is similar to the original functioning of the Metronome clock at Union Square in New York City. It add several additional functions including colored LED lighting, and optional date and weather display.

All settings are stored in non-volatile memory (NVS) and survive power cycles. Configuration is performed through a WiFi access point setup page or a live web interface accessible from any device on the same network.

2. Models

Several models are covered by this manual. The model is set at the factory and cannot be changed by the user. The Zev-7S, Zev-7 Mini, Zev-7-18 and Zev-7RZ are Nixie-tube clocks and share one firmware; the Zev-7NS and Zev-7ND are the equivalent numitron-tube clocks and run their own; the Metronome runs a third. All are operationally similar, differing where the display hardware and its capabilities differ.

The Zev-7RZ uses Dalibor Farny R/Z568M Nixie tubes in place of the standard tubes fitted to the other models. It is otherwise identical to the Zev-7S, and every feature described in this manual applies to it unchanged.

> **NOTICE:** If the display shows 00, the processor must be returned to the manufacturer for reprogramming.

Numitron Variants (Zev-7NS and Zev-7ND)

The numitron models run their own firmware, functionally similar to the Zev-7's, and share every feature described in this manual — Rules, LED effects, colon display patterns, the info display cycle, sensors, OTA updates, and the web interface. Wherever this manual says "Nixie" (for example, "Nixie Brightness"), read it as the numitron display on these models. The differences are:

- Display technology: seven numitron seven-segment filament tubes — DR2010 on the Zev-7NS, DTF-104B on the Zev-7ND — driven by TLC59116 constant-current LED drivers over I2C, in place of the Zev-7's IN-14 Nixie tubes and HV5530 high-voltage shift drivers. The "Nixie Brightness" setting controls numitron filament brightness on these models.
- No high voltage: the numitron display runs at low voltage with no mercury and no ~170 V supply. The high-voltage safety note in the Copyright and Disclaimer section applies to the Nixie models only.
- No cathode anti-poisoning: filaments do not poison, so Section 14 (Cathode Anti-Poisoning), the Anti-Poison settings on the AP and live forms, the power-on conditioning runs, and the noap rule keyword have no effect on the numitron models. The display simply runs the time continuously.
- Colon indicators: the colon separators are LEDs rather than neon lamps, but every colon (neon) display pattern in Section 11 works identically. Because LEDs read much brighter than the filament tubes, a Colon LED Brightness control (0–10) is provided on both the AP Settings form and the live form. It is a trim, not an absolute level: the colons still follow the Nixie Brightness setting and the light sensor, and this control only sets how bright they sit against the tubes. The default is 3. Raise it if the colons look too dim at low display brightness, lower it if they overpower the tubes.
- Models and addressing: Zev-7NS (DR2010 tubes, with a separate under-board light string) and Zev-7ND (DTF-104B tubes, with no under-board string — the Under Lighting settings are hidden

on that model). In AP mode they appear as the WiFi networks Zev-7NS and Zev-7ND, and are reachable by name at zev-7ns.local and zev-7nd.local respectively.

3. Display

The Zev-7 has seven Nixie tubes; the Metronome has fifteen. Both have a set of neon indicator lamps:

Position	Tube	Shows
Left	NX1	Hours tens
	NX2	Hours ones
	NX3	Minutes tens
	NX4	Minutes ones
	NX5	Seconds tens
	NX6	Seconds ones
Right	NX7	Tenths of a second

Neon indicators between the tube pairs serve as colon separators (H:M and M:S) and status indicators. A separate "BR" neon is used for additional indication and blinks off briefly if the clock loses its WiFi connection.

The tenths-of-a-second digit updates live at 10 Hz.

The Metronome display adds an additional eight tubes. NX8 shows live hundredths of a second; the remaining seven count down the time to midnight.

Position	Tube	Shows
Center	NX8	Hundredths of a second
	NX9	Tenths of a second
	NX10	Seconds ones
	NX11	Seconds tens
	NX12	Minutes ones
	NX13	Minutes tens
	NX14	Hours ones
Right	NX15	Hours tens

4. Buttons

The clock has two buttons: SW1 (left) and SW2 (right).

SW1 — During Normal Operation

Action	Function
Single click	Wake the display when an active rule has dimmed or turned it off (Rules Timeout)
Double-click	Scroll IP address across the tubes
Hold 10 seconds	Start OTA firmware update

SW2 — During Normal Operation

Action	Function
Single click	Cycle Nixie brightness 1–9 (temporary, not saved to memory)
Triple-click	Enter light sensor calibration mode (~30 seconds)
Hold 5 seconds	Restart in AP configuration mode

Button Override

Pressing SW1 temporarily cancels any active rule (for example, a rule that has dimmed or turned off the display overnight) and restores the standard display brightness, LED, and underlight settings. The wake lasts for the configured Rules Timeout (15 Seconds, 1 Minute, or 5 Minutes). When the timeout expires, the active rule re-applies.

5. Power-On Functions

On the numitron models the SW1-only and SW2-only power-on functions below do nothing (there are no cathodes to condition); only the SW1 + SW2 factory reset applies.

Special functions are available by holding buttons while applying power. These are read before the clock initializes its normal button handling.

Buttons Held	Function
SW1 only	Startup display test and cathode exercise (5 min). Cycles all digits 0–9 on all tubes at full brightness.
SW2 only	Anti-poison conditioning run (5 min). Rapidly cycles all cathodes to recondition tubes.
SW1 + SW2	Factory reset. Clears all saved settings and restarts into AP configuration mode.

6. AP Configuration Mode

Entering AP Mode

- Hold SW2 for 5 seconds during normal operation, OR
- Apply power with SW2 held down

AP mode is indicated by a single lit tube that bounces left and right across NX1 through NX7 (Zev-7) or NX1 through NX15 (Metronome), with the digit counting 0–9 as it bounces.

Connecting

1. Connect a phone or computer to the WiFi network named for your model — Zev-7S, Zev-7M, Zev-7-18, Zev-7RZ, Zev-7NS, Zev-7ND or Metronome — or the custom AP name.
2. Open a web browser and go to 192.168.4.1
3. Configure settings and press Save. Press Back, and press Start.

The setup form is split across multiple pages (Network, Display, Time Zone, Sensors, Rule Editor, etc.). Use the page buttons on the home page to navigate.

Network Settings

Setting	Description
Access Point Name	WiFi network name when in AP mode (default: the name of your model)
SSID	Select the home WiFi network from the scan list
SSID Override	Manually type an SSID if it doesn't appear in the scan
Password	WiFi password
NTP Server	Primary time server (default: time.cloudflare.com)
Alternate NTP Server	Backup time server, used if the primary becomes unreachable (default: time.google.com)
Live Form Password	Optional password for the live web interface. Leave blank for no password.
Live Form Enabled	Turns the live web interface on or off (default: On)

Display Settings

Setting	Options	Default

Time Format	12 Hour, 24 Hour	12 Hour
Date Display Format	Off, MM/DD/YY, DD/MM/YY, YY/MM/DD	MM/DD/YY
Info Transition	Fade, Scroll, Wave, Spin Wave	Spin Wave
Neon Display	See Section 11: Neon Display Modes	Railroad
Nixie Brightness	1–9	8
LED Effect	See Section 9: LED Effects	Random Multi
RGB LED Brightness	0–9	4
LED Follows Digit	On, Off (LED under each tube reflects that tube's digit)	Off

Digit Change Effect

Option	Description
Off	Instant digit change, no animation
Crossfade	Smooth PWM fade between old and new digits
Slot Machine 10s	Cascading countdown effect every 10 seconds
Slot Machine 60s	Cascading countdown at the minute boundary with ease-out timing

Crossfade Duration is independently configurable from 0 (instant) to 9 (900 ms) in 100 ms steps. Default is 5 (500 ms). Slot Machine Speed is configurable 0–9 (default 5); higher values run the cascade faster.

Anti-Poison Settings (Nixie models only)

Setting	Options	Default
Anti-Poison Cycle	Off, 5 Seconds Every Minute, 10 Seconds Every Minute, 1 Minute Every 10 Minutes, 1 Minute Every Hour, 10 Minutes Every Hour, 30 Minutes at Midnight	5 Seconds Every Minute
Start Second	0–59	30

Rules Timeout

How long the SW1 button-wake lasts when a rule has dimmed or turned the display off: 15 Seconds, 1 Minute, or 5 Minutes (default: 15 Seconds). When the timeout expires, the active rule re-applies.

Time Zone

Setting	Options	Default
UTC Offset Hours	0–23	5
UTC Offset Minutes	0–59	0
UTC Offset Direction	Minus, Plus	Minus

Daylight Saving Time

Setting	Options	Default
DST Start Month	Jan–Dec	March
DST Start Week	1–5	2
DST Start Day	Sun–Sat	Sunday
DST Start Hour	0–23	2
DST Offset Hours	0–13	1
DST Offset Direction	Plus, Minus	Plus

Standard Time

Setting	Options	Default
STD Start Month	Jan–Dec	November
STD Start Week	1–5	1
STD Start Day	Sun–Sat	Sunday
STD Start Hour	0–23	2

Default DST/STD settings are configured for US Eastern Time.

Weather Sensor (BME280)

Setting	Options	Default
Temperature Display	Off, Fahrenheit, Celsius, Kelvin	Fahrenheit
Temperature Offset	−15 to +15 (1° increments)	0
Display Pressure	Off, Inches Mercury, Millibars	Off
Display Humidity	Off, Yes	Off

Light Sensor

Setting	Options	Default
Light Level Threshold	Off, 1–50	Off

Minimum Nixie Brightness	0–9	4
Minimum LED Brightness	0–9	4

Motion Sensor (PIR)

Setting	Options	Default
Running Timeout	Off, 10s, 20s, 30s, 1 min, 5 min, 10 min, 15 min, 30 min, 1 hr	Off
Rules Timeout	Off, 10s, 20s, 30s, 1 min, 5 min, 10 min, 15 min, 30 min, 1 hr	Off

Under Light

Setting	Options	Default
Brightness	0–9	4
Color	RGB color picker	Blue (0000FF)

Miscellaneous

Setting	Options	Default
Date Display Start Second	0–59	20
Weather Display Start Second	0–59	40
Firmware Update URL	URL	https://mitchsf.github.io/Firmware/

The Date Display and Weather Display sections each have a Start Second setting, and the two run independently. Keep them several seconds apart so a long info cycle on one doesn't collide with the other. The Weather Display Frequency setting controls how often the weather cycle runs — from Every Minute (the default) down to Hourly. Every choice divides evenly into an hour, so the spacing stays even across the top of the hour.

7. Live Settings Web Interface

Once connected to WiFi, the live settings page is accessible from any device on the same network.

Browse to http://<clock-ip-address>/   To find the IP address, double-click SW1 — the address scrolls across the tubes.

Using a Name Instead of an IP Address (mDNS)

On many networks the clock can be reached by name instead of by IP address. Browse to http://zev-7.local/ (or http://zev-7m.local/ for the Mini, http://zev-7-18.local/ for the Zev-7-18, http://zev-7rz.local/ for the Zev-7RZ, http://zev-7ns.local/ or http://zev-7nd.local/ for the numitron models, or http://metronome.local/ for the Metronome) and the page should load. The hostname is derived from the Access Point Name set during configuration, lowercased, with any spaces or special characters converted to hyphens. For example, an AP Name of "Zev-7" becomes zev-7.local; "My Office" becomes my-office.local.

If the Access Point Name has been customized during AP configuration, the mDNS address follows the same rule: the new AP Name (lowercased, with any non-alphanumeric characters converted to hyphens) followed by ".local". This makes it easy to give each clock in a multi-clock environment a unique address — for instance, changing the AP Name to "Kitchen Clock" on one unit makes that clock reachable at http://kitchen-clock.local/ regardless of its IP.

This feature relies on mDNS (also known as Bonjour or zero-configuration networking) and does not work on every network. Some routers — particularly older models or routers with multicast filtering enabled — block the discovery messages mDNS depends on, in which case the .local address will not resolve. Some browsers and operating systems also handle mDNS inconsistently: macOS, iOS, and Windows 10/11 generally work out of the box, while some Android versions and Linux distributions require additional setup. If the .local address does not load, fall back to the IP address (double-click SW1 to display it).

If a Live Form Password was set during AP configuration, the page prompts for it before any settings are shown.

Install as a Home-Screen App (Add to Home Screen)

The live web interface can be added to a phone or tablet home screen so it launches full-screen like an app, showing the clock's own nixie-tube icon and name. From the live home page, open the System page and tap Add to Home Screen. On Android this offers to install the page directly; on iPhone and iPad it reminds you to use Safari's Share button and choose "Add to Home Screen" (Apple requires that step to be done from the browser's Share menu). Either way, an icon appears on the home screen that opens straight to the clock's settings.

The icon is labeled with the clock's Access Point Name, so in a multi-clock home each unit gets its own recognizable icon — for example "Kitchen Clock."

A home-screen icon remembers whichever address was open when it was created. For an icon that keeps working, add it from the clock’s name address (http://<name>.local/, see above) or give the clock a fixed IP address, so a changing DHCP lease cannot leave the icon pointing at the wrong place. If an icon ever stops opening, delete it and add it again from a working address.

Available Controls

All changes take effect immediately without rebooting. Use the Save to Memory button to persist changes across power cycles. The form is split across multiple pages. The address bar shows which page is currently displayed. Some settings are only available on the Settings pages. These are accessed by holding SW2 for five seconds.

- Nixie Brightness (1–9)
- RGB LED Brightness (0–9)
- Colon LED Brightness (0–10, numitron models only)
- Neon Display (all 9 patterns)
- LED Effect (all 13 effects); LED Color picker (active when LED Effect is Custom)
- Digit Change Effect, Crossfade Duration, Slot Machine Speed
- Info Transition (Fade, Scroll, Wave, Spin Wave)
- Date Display (Enable/Disable; format is set on the Settings form)
- Temperature Display, Offset, Pressure, Humidity
- Anti-Poison Cycle, Start Second
- Motion Sensor: Running Timeout, Rules Timeout
- Light Sensor settings
- Under Light Brightness and Color
- LED Follows Digit
- Date Display Start Second; Weather Display Frequency and Start Second
- Rule Editor page (add, edit, and delete time-range rules)
- Disable Rules toggle (temporarily ignores all rules without deleting them)
- Show IP Address button

8. Rules

Rules schedule different display, LED, under-light, and motion-sensor settings for specific times of day or days of the week. Up to 32 rules can be defined. Each rule specifies a time range and one or more overrides — for example, "turn the display off from 11:00 PM to 7:00 AM" or "set the LEDs to red at brightness 9 from 6:00 PM to 10:00 PM on weekdays."

Priority: rules are applied setting by setting, not as whole rules. Where two rules overlap in time, each setting is taken from the rule that sets it, and only when both rules set the SAME setting does the one listed higher in the schedule win. So a rule that dims the tubes and a rule that changes the under-light colour both take full effect during an overlap - they touch different settings and neither cancels the other. Layering a broad rule with a narrower exception above it is the normal way to build a schedule. Use the up and down arrows on the right of each row to change a rule’s priority (up = higher priority). The Rule Editor shows a warning icon (⚠) only where an overlap actually contests a setting; hover or tap it to see which setting and which rule. A rule marked this way is not disabled - everything else in it still applies, including that setting outside the overlapping window.

The Disable Rules toggle on the Rule Editor page temporarily ignores all rules without deleting them. This is useful for making live adjustments without rules re-applying. The toggle does not persist across reboots and always defaults to Off (rules active) when the live page is opened.

Rule Editor page. Tap Rule Editor on the live form home page to manage the rule list. Each rule is written as a short text expression and appears in the schedule table below the input field. Tap a row in the schedule to load that rule into the editor; tap Apply to save it. Tap Delete to remove the loaded rule.

Syntax. A rule starts with a time range (t=...) followed by any combination of field overrides, separated by commas or spaces. Time ranges use a 24-hour clock and the end time is exclusive (t=5-7 covers 5:00 through 6:59). Times may be written as a bare hour (t=5 to 5:00-5:59), an hour range (t=5-7), exact minutes (t=5:30-6:00), or in compact four-digit form (t=0530-0600 = 5:30-6:00). Cross-midnight ranges are allowed: t=22-6 means 10:00 PM through 5:59 AM.

Days of the week. Beyond the wd (weekday) and we (weekend) shortcuts, a rule can target specific days by name — sun, mon, tue, wed, thu, fri, sat, or the full names such as tuesday. A day range like tue-thu covers a span and may wrap the week (sat-mon covers Saturday, Sunday, Monday); a single day name limits the rule to that day. Day terms are combined as a union and may be mixed with wd/we. Separately, the value off may be used in place of 0 on any numeric field — nb=off, lb=off, and so on.

Dark rules. The bare dark keyword makes a rule apply only while the room is dark. It requires the Light Sensor (Section 13) to be enabled. The clock waits for about a minute of continuous darkness before dark rules engage — brief shadows, headlights, or a passing cloud will not trigger them — and releases them within a few seconds once the room is lit again. dark combines with time ranges and day filters; use t=all with dark for a rule that follows the room rather than the clock. A dark rule on a clock whose Light Sensor is disabled stays dormant — it is accepted but never fires.

Sunrise and sunset. Instead of a clock time, either end of a rule's time range can be anchored to today's sunrise or sunset, so a seasonal rule keeps following the sky instead of needing an edit twice a year. t=18:00-7:00 is right in December and two hours wrong in June; t=sunset-sunrise is right all year. Both ends may be anchors, and anchors mix freely with clock times: t=sunset-sunrise, t=sunrise-sunset, t=sunset-23:00, t=23:00-sunrise. Both keywords may be shortened to sr and ss - t=ss-sr is the same rule as t=sunset-sunrise, and the long and short forms may be mixed freely; the schedule table always shows the long form. Everything else about the rule is unchanged - day filters, dark, noap and the full field list all work the same way.

Offsets. An anchor can be shifted by writing + or - followed by a number and a unit: m for minutes, h for hours, up to 12 hours either way. For example t=sunset-30m-23:00 starts 30 minutes before sunset and runs to 23:00, and t=sunrise+1h-sunset-1h covers only the solidly-daylight hours. The unit is required. A bare number after the keyword is read as a time, not an offset - t=sunset-0700 means "sunset until 07:00" - so an offset without its unit is rejected, with a message showing the corrected text.

Location. Sunrise and sunset depend on where the clock is, so set Latitude and Longitude on the System page of the live settings form before using these rules. A map above the two fields lets you tap your location rather than typing it; it starts locked, so uncheck Lock Map first. The map needs an internet connection and is only available on the live form, not the initial AP setup form - the two text fields work either way. A sunrise or sunset rule entered with no location set is refused rather than silently ignored, and clearing the location warns that any such rules are paused.

How the times are found. The clock calculates sunrise and sunset itself, from your latitude, longitude and the date - there is no internet lookup and nothing to subscribe to. It uses the standard NOAA sunrise equation: from the number of days since the year 2000 it finds where the Earth is in its orbit, corrects for the orbit being an ellipse rather than a circle, and from that gets the sun's position for the day - the moment of local solar noon and how far north or south of the equator the sun stands. Combining that with your latitude gives how long the sun takes to travel from overhead to the horizon, which is subtracted from solar noon for sunrise and added for sunset. The horizon is taken as 0.833 degrees below level, which allows for the width of the sun's disc and the way the atmosphere bends its light - the sun has really already set when you watch it touch the horizon. The result is accurate to well under a minute. Times are recalculated each day at midnight, and immediately whenever you change a rule or the location. At extreme latitudes, on days when the sun never rises or never sets, sunrise and sunset rules simply do not fire.

Note: lc is honored only when le=13 (Custom). Using lc by itself auto-promotes the rule to effect 13. Pairing lc with any other le value is rejected by the rule editor. (Effect numbering shifted when Chroma Digit was added: Cycle is 12 and Custom is 13.)

Keyword Reference

Keyword reference. Each rule may use any combination of the following keys. An override is written as a key, an equals sign and a value. Short and long forms are interchangeable, and fields may appear in any order; the time range does not have to come first.

Keyword	Description
t / time	Time range. Specifies when the rule is active. Required in every rule. Format: HH:MM-HH:MM, H-H, HHMM-HHMM, or bare H. End time is exclusive, and a range may cross midnight. Either end may instead be sunrise or sunset, optionally offset by minutes or hours (sunset-30m, sunrise+1h) — see Section 8. Sun anchors need Latitude and Longitude set on the System page.
nb / nixiebrightness	Nixie tube brightness (0-9, 0 = display off).
cd / colondisplay	Colon neon pattern (0-8). See Section 11 for the pattern catalog.
lb / ledbrightness	Digit RGB LED brightness (0-9, 0 = LEDs off).
le / ledeffect	Digit LED effect (0-13). See Section 9 for the effect catalog. Index 13 (Custom) renders a solid color from the lc value below; index 12 (Cycle) walks the effect order list; any other index runs the named animation.
lc / ledcolor	Digit LED solid color. Hex (#RRGGBB) or color name (red, green, blue, white, yellow, cyan, magenta, orange, purple, pink, black). Honored only when le=13 (Custom). Using lc by itself auto-promotes the rule to effect 13; pairing lc with any other le value is rejected by the rule editor.
ub / underbrightness	Under-light brightness (0-9, 0 = off). No effect on the Zev-7ND, which has no under-board string.
uc / undercolor	Under-light color. Same hex or color-name format as lc. No effect on the Zev-7ND.
pir / passiveinfrared	Motion-sensor timeout (0 = Off, 1 = 10s, 2 = 20s, 3 = 30s, 4 = 1m, 5 = 5m, 6 = 10m, 7 = 15m, 8 = 30m, 9 = 1hr).
wd / weekday	Bare keyword (no = value). Restricts the rule to weekdays (Monday-Friday).
we / weekend	Bare keyword (no = value). Restricts the rule to weekends (Saturday-Sunday).
noap / noantipoison	Bare keyword (no = value). Suppresses anti-poison cycles during this rule's window. Useful for quiet/dim windows where a full-brightness anti-poison run would be disruptive. Nixie models only; ignored on Zev-7N numitron models.
dark	Bare keyword (no = value). The rule applies only while the room is dark, as measured by the light sensor. Engages after about a minute of continuous darkness; releases within a few seconds when the room lights up. Requires the Light Sensor to be enabled.
off	Bare keyword (no = value). Shortcut that expands to nb=0, lb=0, ub=0 (everything off: Nixie tubes, digit LEDs, and under-light). On the Zev-7ND, ub is accepted but has no effect.
Day names / ranges	Bare day tokens restrict a rule to specific days of the week: sun, mon, tue, wed, thu, fri, sat (full names such as tuesday also work). A range spans consecutive days and may wrap the week — tue-thu means Tuesday through Thursday; sat-mon means Saturday, Sunday, Monday. A single token means that one day only. Day terms combine as a union and may be mixed with wd/we — for example wd, sat covers Monday through Saturday.
he / heal	Marks the time range as a Tube Healing window and sets the session length in days (1–99; a bare heal means 30 days; heal=0 cancels healing inside a higher-priority window). During a healing window the cathodes selected on

	the Tube Healing page glow to recover from poisoning while every other tube stays dark, and the motion sensor is held off for the window. See Section 15. The tubes run at full brightness for the window regardless of the brightness setting, an nb value in another rule, or the light sensor. Nixie models only.
off (as a value)	On any numeric field, off may be typed in place of 0 — for example nb=off, lb=off, ub=off, or heal=off. This is separate from the bare off shortcut above, which sets nb, lb, and ub together.
sunrise / sunset (sr / ss)	Used in place of a clock time at either end of the time range, so the rule follows the season. May be written in full or as the two-letter short forms sr and ss - t=ss-sr is the same rule as t=sunset-sunrise, and the two forms may be mixed. The schedule table always shows the long form. Add an offset with + or - plus a number and a unit - m for minutes, h for hours, up to 12 hours (t=ss-30m-23:00). The unit is required: a bare number is read as a time, so t=sunset-0700 means "sunset until 07:00". Requires Latitude and Longitude on the System page.

Examples

Rule	Time Format	What It Does
`t=5, nb=9`	H - bare hour	At 5:00 AM, set Nixie brightness to maximum. The bare-hour form covers the full 60 minutes (5:00 through 5:59).
`t=22-6, nb=0`	H-H - cross-midnight hour range	Overnight blackout, every day: turns the Nixie display off from 10:00 PM through 5:59 AM. The end time wraps past midnight automatically.
`t=22:00-06:00, wd, nb=0`	HH:MM-HH:MM - full precision	Weeknight blackout (Mon-Fri only). Same window as the rule above, written with full HH:MM digits and restricted to weekdays.
`t=22-9, we, nb=0`	H-H - cross-midnight + weekend filter	Weekend sleep-in (Sat-Sun only): display off from 10:00 PM through 8:59 AM the next morning. Pairs naturally with the weeknight rule above.
`t=5:30-6:00, lb=9, le=1`	H:M-H:M - minute precision	Sunrise wake: 30 minutes of Red-White-Blue Wave (effect 1) at LED brightness 9 starting at 5:30 AM.
`t=0530-0600, lb=9, le=1`	HHMM-HHMM - compact 4-digit	Identical window to the rule above, written in the no-colon four-digit compact form. Useful when typing on a phone keyboard.

`t=17:30-18:30, lb=9, ub=9, uc=orange`	HH:MM-HH:MM - one-hour minute span	Dinner spotlight, every day: maximum LED and under-light brightness with an orange under-light color, from 5:30 PM through 6:29 PM.
`t=12-13, lc=green`	H-H - single-hour range	Lunch-hour green: tints all digit LEDs solid green from 12:00 PM through 12:59 PM. lc renders a solid color (no effect needed).
`t=6-22, pir=0`	H-H - daytime range	Disables the motion sensor during the day (6:00 AM through 9:59 PM) so the display stays on regardless of motion.
`t=7-9, wd, lb=9, le=1`	H-H + weekday filter	Weekday morning energizer (Mon-Fri only): Red-White-Blue Wave at maximum LED brightness from 7:00 AM through 8:59 AM.
`t=20-23, we, le=5, lb=9`	H-H + weekend filter	Weekend evening (Sat-Sun only): Random Big Color Wave (effect 5) at full LED brightness from 8:00 PM through 10:59 PM.
`t=23:00-23:45, nb=1`	HH:MM-HH:MM - partial hour	Final-stretch dim: minimum non-zero Nixie brightness for the last 45 minutes before midnight.
`time=22:00-06:00, weekday, nixiebrightness=0`	Full keyword names	Identical to the weeknight blackout shown earlier, but written with the full descriptive keyword names instead of the short keys. The parser accepts either form interchangeably.
`nixiebrightness=2, weekend, time=22:00-09:00`	Order shuffled - time is not first	Weekend overnight dim: Nixie brightness drops to 2 from 10:00 PM through 8:59 AM, Saturday and Sunday only. Fields may appear in any order - the time range does not have to come first.
t=7-9, tue-thu, nb=9	Day range	Full Nixie brightness from 7:00 through 8:59 AM, but only on Tuesday, Wednesday, and Thursday. Day names and ranges may be combined and mixed with wd/we.
t=0:00-6:30, heal=30	Healing window	Runs a 30-day Tube Healing session every night from midnight through 6:29 AM: the digits selected on the Tube

		Healing page glow to recover poisoned cathodes while the rest of the display is dark. See Section 15.
t=all, dark, nb=1, lb=0	Whenever dark	Follows the room instead of the clock: whenever the room is dark, tube brightness drops to 1 and the digit LEDs turn off. Everything returns a few seconds after the lights come on.
t=sunset-sunrise, nb=3, lb=2	Dusk to dawn	Dims the tubes to 3 and the LEDs to 2 from sunset until sunrise, following the season automatically — no need to adjust the times as the nights get longer or shorter.
t=sunset-30m-23:00, nb=6, ub=3	Evening, sun-anchored start	Eases the display down half an hour before sunset and holds that until a fixed 23:00 bedtime. Mixes a sun anchor with an ordinary clock time.
t=sunrise+1h-sunset-1h, nb=9	Full daylight only	Runs the tubes at full brightness only during solidly-daylight hours, avoiding the dim half-hour at each end of the day.

9. LED Effects

The Nixie tubes are backlit by RGB LEDs, on the Metronome as well as the Zev-7. Every model except the Zev-7ND also has separate under-light LEDs along the front edge of the base. The under-light LEDs have independent brightness and color control and are not affected by LED effects.

LED effects apply to the tube backlight LEDs:

#	Effect	Description
0	Off	All LEDs dark
1	Red-White-Blue Wave	Red, white, and blue cycling with sine-wave modulation
2	Blue-White Wave	Blue and white alternating wave
3	Red-White Wave	Red and white alternating wave
4	Random All Color Wave	Single random color with sine-wave modulation
5	Random Big Color Wave	Random colors per LED with sine-wave modulation (default)
6	Random Bright Color	Bright saturated random colors with sine-wave modulation
7	Red Wave	Red with sine-wave fade
8	Green Wave	Green with sine-wave fade
9	Blue Wave	Blue with sine-wave fade
10	Larson Scanner	Knight Rider effect cycling through 7 colors
11	Cycle All	Rotates through effects 1–10, changing every 10 seconds
12	Custom	Renders the color set in the LED Color picker (or by the lc rule keyword) as a solid across all LEDs

10. Digit Change Effects

When a time digit changes, the clock can animate the transition:

Effect	Description
Off	Instant change, no animation
Crossfade	Smooth PWM fade from old digit to new digit. Duration configurable 0–900 ms in 100 ms steps (default 700 ms).
Slot Machine 10s	Cascading countdown effect triggered every 10 seconds
Slot Machine 60s	Cascading countdown at the minute boundary with ease-out timing

Slot Machine Speed (0–9, default 5) tunes how fast the cascade runs. 0 makes the change instant; higher values run the cascade faster. Crossfade speed can also be changed in the same way.

11. Neon (Colon) Display Modes

The neon indicator lamps between the tubes can display various patterns:

#	Mode	Description
0	Off	All neons dark
1	All On	All neons continuously lit
2	Bottom Flash	Bottom colon neons blink each second
3	All Flash	All neons blink each second
4	Railroad	Alternating railroad-crossing pattern (default)
5	Strobe	Horizontal stripe alternating pattern
6	Left to Right	Sweep pattern across neon positions
7	AM/PM Top Off	AM/PM indicator with top neons off
8	AM/PM Top On	AM/PM indicator with top neons on

12. Info Display Cycle

The clock periodically interrupts the time display to show additional information. Date and weather information run on independent schedules, each firing at its configured Start Second. The date cycle runs once per minute. The weather cycle runs at the configured Weather Display Frequency: Every Minute (the default), every 2, 3, 5, 10, 15, 20, or 30 minutes, or Hourly. Every interval divides evenly into an hour, so the cycle lands on predictable minutes — Hourly fires at minute :00, Every 30 Minutes at :00 and :30, and so on.

Information Shown (if enabled)

Date: Displayed as MM/DD/YY, DD/MM/YY, or YY/MM/DD depending on configuration.

Temperature: Shown in tenths of a degree (e.g., 78.2). Temperature units selectable as Fahrenheit, Celsius, or Kelvin. Negative temperatures are indicated by a special neon pattern.

Barometric Pressure: Displayed in inches of mercury or millibars.

Humidity: Displayed as a percentage with tenths (e.g., 55.2%).

Each item is held on the display briefly, then transitions to the next using the configured Info Transition (Fade, Scroll, Wave, or Spin Wave). The first info trigger after boot is suppressed so the clock has a chance to settle.

13. Sensors

Temperature / Pressure / Humidity (BME280)

If a BME280 sensor is installed, the clock can display temperature, barometric pressure, and humidity as part of the info display cycle. A calibration offset is available for temperature.

The sensor is auto-detected at startup on the I2C bus (addresses 0x76 and 0x77).

Light Sensor

Calibration: The clock automatically dims its display to match the ambient light in the room, and you can teach it the brightness range of your location. Triple-click SW2 to enter calibration mode. The clock runs a 30-second calibration during which the four left-hand tubes show a live sensor reading (0–4095; note that brighter light produces a lower number). While calibration is running, expose the sensor to the brightest light at which you want the display at full brightness — for example, turn on the room lights or open the blinds — then cover or shade the sensor to register the darkest condition at which you want it dimmed to minimum. The clock tracks the brightest and darkest readings it sees during the 30 seconds; there is no button to press to finish. When the timer expires, calibration ends automatically, the new range is saved to memory, and the clock returns to showing the time. If too small a light range is captured, the clock keeps its previous calibration unchanged. The calibration is stored permanently and survives power loss, so it only needs to be repeated if the clock is moved to a location with noticeably different lighting. The light sensor also drives dark rules — rules that apply only while the room is dark (see Section 8).

Motion Sensor (PIR)

When enabled, the display goes dark after the configured timeout with no detected motion. When motion is detected, the display smoothly fades in to the configured brightness. When motion stops and the timeout expires, the display smoothly fades out.

Two separate timeout settings are available: Running Timeout controls PIR behavior during normal operation, and Rules Timeout controls PIR behavior while a rule has turned the display off. Each can be configured independently. If Rules Timeout is Off (default), the display remains in its rule-driven state regardless of motion. Individual rules can also set their own per-rule PIR timeout with the pir= field (see Section 8).

14. Cathode Anti-Poisoning

Applies to Nixie models only. The Zev-7NS and Zev-7ND numitron variants use filament seven-segment tubes that do not suffer cathode poisoning. On those models this entire section, the Anti-Poison settings, the power-on conditioning runs, and the noap rule keyword have no effect.

Nixie tubes can develop "cathode poisoning" when certain digits are displayed for long periods without exercising other cathodes. The unused cathodes accumulate a chemical deposit that gradually reduces their brightness and produces a faded or "ghosted" appearance when those digits are eventually shown. The anti-poison feature periodically cycles all 10 digits on every tube at high speed to prevent and reverse this effect.

Anti-Poison Cycle is a single setting that picks a paired duration and schedule. The available cycles are 5 Seconds Every Minute, 10 Seconds Every Minute, 1 Minute Every 10 Minutes, 1 Minute Every Hour, 10 Minutes Every Hour, and 30 Minutes at Midnight.

Trigger conditions: all cycles fire at the configured Start Second. The five non-Midnight cycles only run when the tubes are currently on — if an active rule has the display off, or the PIR motion timeout has blanked it, the cycle is skipped that interval. The Midnight cycle is the exception: it always runs, even when the display is otherwise off.

During a cycle: tubes run at full brightness regardless of any active rule. RGB LEDs continue running their configured effect and brightness. With LED Follows Digit enabled, the under-tube LEDs follow whichever tubes are lit — since the cycle keeps all tubes lit, the LEDs stay on through the cycle. Both colon neons are lit. If PIR turns the tubes off mid-run on a non-Midnight cycle, the cycle aborts cleanly and the display goes back to its off state.

Smart conditioning algorithm: rather than spending an equal amount of time on every digit, the firmware tracks which cathodes get exercised by the normal time display and weights the anti-poison cycle to favor the ones that do not. For example, in 12-hour mode the leftmost hour tube only ever displays a "1", so digits 0 and 2 through 9 on that tube get roughly four times the conditioning time of digit 1. The minute and second tens tubes never display 6 through 9 during normal use, so those digits also get extra time. The algorithm uses separate weighting tables for 12-hour and 24-hour modes, so changing the time format automatically rebalances the cycle.

The Anti-Poison Start Second setting controls when in the minute the cycle begins. This can be offset from the Date and Weather Display start seconds to avoid conflicts. When running multiple clocks, different start seconds can synchronize or stagger the cycles.

Manual Conditioning (Hold SW2 at Power-On)

A separate, more aggressive 5-minute conditioning cycle is available as a power-on function. This is intended for tubes that are visibly faded, have been in storage for a long time, or have not been exercised by the regular anti-poison schedule.

To start a manual conditioning run:

1. Remove power from the clock.
2. Press and hold SW2 (right button).
3. Apply power while continuing to hold SW2.
4. Release SW2 once the conditioning cycle begins.

All seven tubes will rapidly cycle through every digit 0–9 at full brightness for 5 minutes. Both colon neons remain lit throughout the run. The clock does not connect to WiFi or fetch the time during this period — it focuses entirely on conditioning the cathodes.

After 5 minutes the cycle ends automatically: the display clears, the neons turn off, and the clock proceeds with its normal boot sequence (WiFi association, NTP sync, then time display). Power can be removed at any point to stop the cycle early; no settings are changed by this operation.

Recommended use: run manual conditioning when first installing the clock, after the clock has been unplugged for several weeks, or any time the tubes show visibly faded digits that the scheduled anti-poison has not corrected. There is no harm in running it more than once.

15. Tube Healing

Anti-poisoning (Section 14) prevents cathode poisoning; Tube Healing recovers a cathode that is already poisoned. If a particular digit on a particular tube looks faded or ghosted even after the regular anti-poison schedule has been running, Tube Healing lets you glow just that cathode for an extended period to burn the deposit off. Like anti-poisoning, this is a Nixie-only feature: the numitron models use filament tubes that do not poison, and have no Tube Healing page.

Where it is. Tube Healing lives on the live web interface, not the AP setup form. From the live home page, open the System page and tap Tube Healing. The page has three parts: a Healing on/off switch, a row of digit-test buttons (0–9), and a grid of checkboxes with one column per digit (0–9 across the top) and one row per tube (numbered from the left).

1. Find the weak cathodes. Tap a digit-test button and that digit is shown on every tube for a few seconds while the button turns blue. Step through the digits and note which tube-and-digit combinations look dim or patchy compared with the rest.

2. Select and arm. In the grid, tick each weak digit in its tube's row (tube 1 is the leftmost). Turn the Healing switch on. Selections are saved immediately and survive a reboot.

3. Schedule when it runs. Healing only treats during a rule window that carries the heal keyword (see Section 8). Add a rule such as t=0:00-6:30, heal=30 to heal every night from midnight to 6:30 AM for a 30-day session. A bare heal means 30 days; heal=N sets any length from 1 to 99 days. An overnight window keeps the clock readable during the day.

What you see during a window. The clock stops showing the time entirely: only the selected cathodes glow, on their own tubes, and every other tube is dark. This is deliberate — a healing digit held in place of a time digit would be unreadable and would dilute the treatment. If more than one digit is selected on the same tube, that tube rotates through them, five minutes each. The rotation begins when the window opens, starting with the lowest selected digit, so every digit — including the first — gets its full five minutes; a window that ends and later reopens starts the rotation over from the first selected digit. The motion sensor, the scheduled anti-poison cycles, and the date and weather info displays are all suspended for the duration of the window so the treatment runs uninterrupted. The tubes run at full brightness for the whole window, regardless of the brightness setting, an nb value in any rule, the light sensor, or the motion sensor — the glow itself is the treatment, so a dimmed cathode heals slowly or not at all. Every tube with a digit selected takes part, including the tenths tube, even if it is normally switched off. Outside the window the clock behaves normally, with the brightness setting and any rules back in charge.

When it ends. The session ends automatically after the number of days set by heal=N (30 by default); the Healing switch then turns itself off, while the digit selections stay ticked so a follow-up session only needs the switch turned back on again. Turning the switch off at any time ends the session immediately. Light poisoning typically clears in about 30 days; heavy poisoning may need a second session. Note: a faded glow only along the bottom of a digit usually indicates a tube air leak rather than poisoning, and cannot be healed.

16. OTA Firmware Update

The clock can update its firmware over WiFi. To trigger an update, hold SW1 for 10 seconds.

The firmware can also be updated from the web settings page, without using the button. On the System page, press Update Firmware. The settings form is replaced by a status message while the clock checks the firmware server. If the firmware is already current, the message says so (with the version number) and the page returns to the home screen automatically. If a newer version is available, the message changes to “updating to v<new version>” and stays on screen while the clock downloads, installs, and reboots; once the clock is back up, the page returns to the home screen on its own. If the update fails, the clock reboots unchanged and the page also returns home — check the tubes for the error code below. The tubes show the same update status described below regardless of whether the update is started from the button or the web page.

Display Feedback

All OTA status appears on the leftmost tubes: download progress uses NX1–NX3, error codes use NX1–NX2. All other tubes are blank during the update.

Display	Meaning
All tubes blank	Checking the firmware server for a new version
Version number (e.g., 2 4 3)	Firmware is already current. Displayed for 5 seconds, then the clock returns to normal.
0	New version found — the download is about to begin
0–100	Downloading and installing (percent complete). 100 holds for 5 seconds, then the clock reboots into the new firmware.

Error Codes

If the update fails, NX1-NX2 display a two-digit error code, then the clock reboots after a few seconds:

Code	Meaning
91	No WiFi connection or firmware URL not configured
92	Could not fetch the VERSION file from the server
93	VERSION file is malformed (missing or invalid version string)
94	Binary download or flash write failed

Version Behavior

Firmware updates never reset your settings. Every update - patch, minor or major - preserves your configuration and WiFi credentials.

Settings are returned to factory defaults only by the deliberate reset in Section 17, or on the very first start-up of a new clock.

Automatic Updates

The clock can keep itself up to date. Automatic Updates, on the System page of the live settings form, is on by default. Once a week, at a time between 2:30 and 3:30 in the morning that is fixed for your particular clock, it checks whether a new release has been published and approved, and installs it if so.

These updates are silent. The display blanks for the download rather than showing progress digits, and the clock restarts without its usual version display or IP scroll - it simply shows the time again a minute or so later, with every setting intact. If anything goes wrong, no matter what, the update is abandoned quietly and retried at the next weekly check; nothing is shown and the clock keeps running the firmware it already had.

Not every release is installed automatically. A release is only picked up once it has been marked as approved for unattended installation, so a new version can be published for manual updating first and approved later. Turning Automatic Updates off leaves the clock entirely under your control; the Update Firmware button and the SW1 hold both continue to work exactly as described above.

Automatic Recovery from a Failed Update

An update that transfers and installs correctly could still, in principle, contain a fault that stops the clock starting up. With updates able to install unattended overnight, that would be the worst possible outcome: a dark clock in the morning, with no display, no settings page and no way to send it new firmware short of connecting it to a computer.

The clock guards against this by keeping the previous firmware. A newly installed version is treated as provisional and has to prove itself by completing a full, successful start-up - reaching the point where the display is running and the clock is on the network. Only then is it accepted permanently. If it fails or restarts before getting that far, the clock discards it on the next restart and goes back to the firmware it was running before, with all settings intact. There is nothing to do and nothing to press; the clock comes back on its own.

One consequence is worth knowing. Because the checkpoint comes after the clock joins your WiFi network and sets its time, an update installed while the network is unavailable can be rolled back even though the firmware itself is sound. If an update seems not to have taken effect, check that the clock is reaching your network and run the update again.

17. Factory Reset

To reset all settings to factory defaults:

1. Remove power from the clock.
2. Press and hold both SW1 and SW2.
3. Apply power while holding both buttons.
4. Continue holding until the display clears (about 5 seconds).
5. Release both buttons. The clock restarts in AP configuration mode.

All WiFi credentials, display preferences, rules, and sensor settings are erased. The model code (which identifies the model) is preserved.

18. REST Programming

Every control on the live settings form can also be operated directly by URL, with no browser form involved. This style of interface is called REST (Representational State Transfer) — the web's convention for controlling things through plain URLs and standard web requests. Each setting is a named field, and a plain web request sets it — the same mechanism the form itself uses. This makes the clock scriptable from anything that can fetch a URL: a browser address bar, curl, a shell script, a phone shortcut, or a home-automation system such as Home Assistant.

Discovering the available fields. Browse to `http://<clock-address>/fields` to get a machine-readable (JSON) list of every controllable field. Each entry shows the field's name, its label as it appears on the form, its type, its current value, and what values it accepts — the minimum/maximum for sliders, or the full option list for dropdowns. For example, an entry like

```
{"name":"brightness","label":"Nixie Brightness","type":"range",
 "page":1,"value":7,"min":1,"max":9,"step":1}
```

says the field `brightness` is a slider currently set to 7 that accepts 1–9. Dropdown entries include an `options` array; the value to send is the option's position in that list, starting at 0 (or, where an `offset` is shown, the position plus that offset). Password fields are never included in the list.

Setting a field. Request `http://<clock-address>/?field=<name>&value=<value>`. The change takes effect immediately, exactly as if the control had been moved on the form. Examples, using the clock's name address (an IP address works the same):

`http://zev-7s.local/?field=brightness&value=3`	Nixie brightness to 3
`http://zev-7s.local/?field=ledbr&value=0`	Digit LEDs off
`http://zev-7s.local/?field=ledmode&value=13`	LED Effect to Custom
`http://zev-7s.local/?field=ledcolor&value=16711680`	LED color to red
`http://zev-7s.local/?field=tempen&value=1`	Temperature display on

Value conventions: toggles take `0` (off) or `1` (on); colors take a single decimal RGB number (red = 255×65536 = 16711680, green = 65280, blue = 255); time fields take HHMM (e.g. `2130` for 9:30 PM); buttons listed with type `action` or `button` are pressed by sending `value=1`.

Saving. Changes made this way are live but temporary — like the form, they last until the next power cycle unless saved. Request `http://<clock-address>/?save=1` to persist everything changed since the last save, the equivalent of pressing Save to Memory.

From the command line or a script:

```
curl "http://zev-7s.local/?field=brightness&value=3"
curl "http://zev-7s.local/?save=1"
```

If a Live Form Password is set, include it as the password of a standard HTTP login (the username is ignored):

```
curl -u :mypassword "http://zev-7s.local/?field=brightness&value=3"
```

Things to know:

- Setting any display field while a rule is active disables the rules, exactly as a change on the form does (see Section 8). For schedules the clock can handle itself — dimming at night, changing colors by time of day — rules are the better tool; REST control is for triggers the clock cannot know about, such as presence detection, sunset lighting scenes, or a "movie mode" button in a home-automation dashboard.

- Field values are applied but not range-checked beyond the form's own limits; stay within the min/max shown by /fields.
- The interface is plain HTTP on the home network. The Live Form Password protects it from casual access, but as with the form itself, do not expose the clock directly to the internet.

19. Troubleshooting

Symptom	Possible Cause	Solution
Display is blank after power on	Waiting for WiFi association or NTP time sync	NX1 scrolls digit "1" while WiFi is connecting and digit "2" while NTP is syncing. If neither completes within 2 minutes, the clock automatically reboots and tries again.
Display shows steady 99	Clock is in AP config mode	Connect to the WiFi network named for your model and configure at 192.168.4.1.
Display shows steady 00 and never starts	Missing or invalid model code	Contact the manufacturer. The unit needs to be re-provisioned.
Tubes are dim	An active rule, or the light sensor adjusting for low light	Press SW1 to temporarily wake the display. Check the active rules on the Rule Editor page and the Light Sensor settings.
BR neon blinks off briefly	WiFi connection lost	The clock is attempting to reconnect automatically. If it persists, check the router or re-enter AP mode.
No WiFi connection at boot	Wrong credentials or out of range	Enter AP mode (hold SW2 for 5 seconds) and reconfigure WiFi.
Time is wrong	Incorrect time zone or DST	Check UTC offset and DST/STD configuration in AP setup.
Time is briefly wrong, then corrects	Primary NTP server slow; alternate took over	Normal behavior. The clock falls back to the alternate NTP server if the primary is stale for more than 30 minutes.
Digits appear faded or ghosted	Cathode poisoning	Enable anti-poison schedule. Hold SW2 at power-on for a 5-minute conditioning run.
OTA shows error 91	No WiFi or missing URL	Ensure WiFi is connected and firmware URL is configured.
LEDs not lighting	Brightness set to 0, or an active rule has dimmed or disabled the LEDs	Check RGB LED Brightness on the live page. Rules can override LED brightness and effect — review the Rule Editor.
Live web interface won't load	Live form disabled or wrong password	Verify Live Form Enabled is On in AP setup. Verify the Live Form Password if one was set.
Under-light not working	Brightness set to 0, or an active	Check Under Light Brightness

	rule	on the live page. Rules can independently set the under-light brightness and color — review the Rule Editor.

Copyright and Disclaimer

The Nixie tubes used in this clock contain a small amount of mercury and neon gas under low pressure. Operate the clock at the supplied voltage only; the high-voltage circuitry inside generates approximately 170 volts and should not be opened or serviced by the end user. This applies to the Nixie models only: the Zev-7NS and Zev-7ND numitron variants use low-voltage filament tubes that contain no mercury and generate no high voltage.

Zev-7 Nixie Clock — Designed by Mitchell Feig d/b/a Angry Electrons